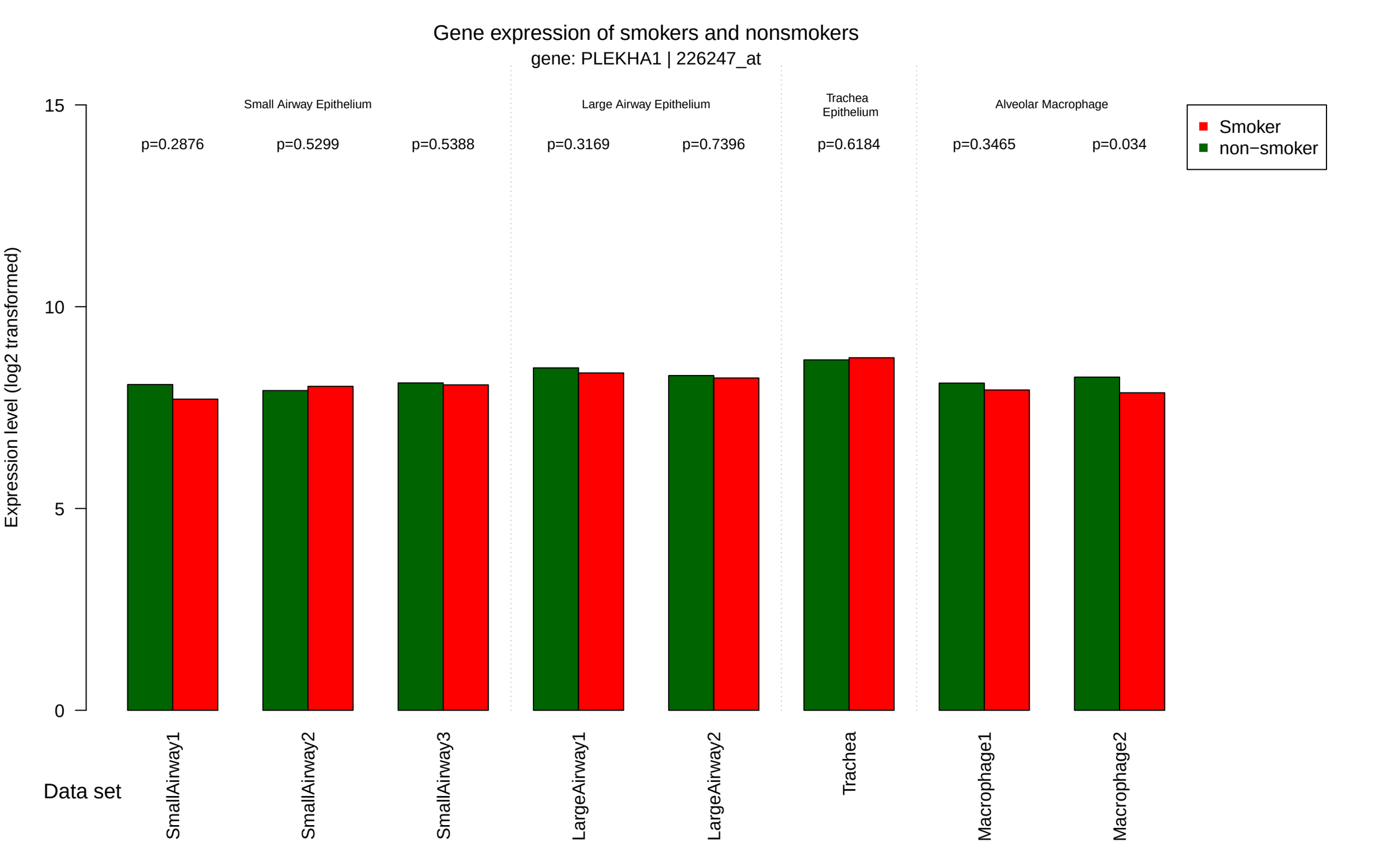
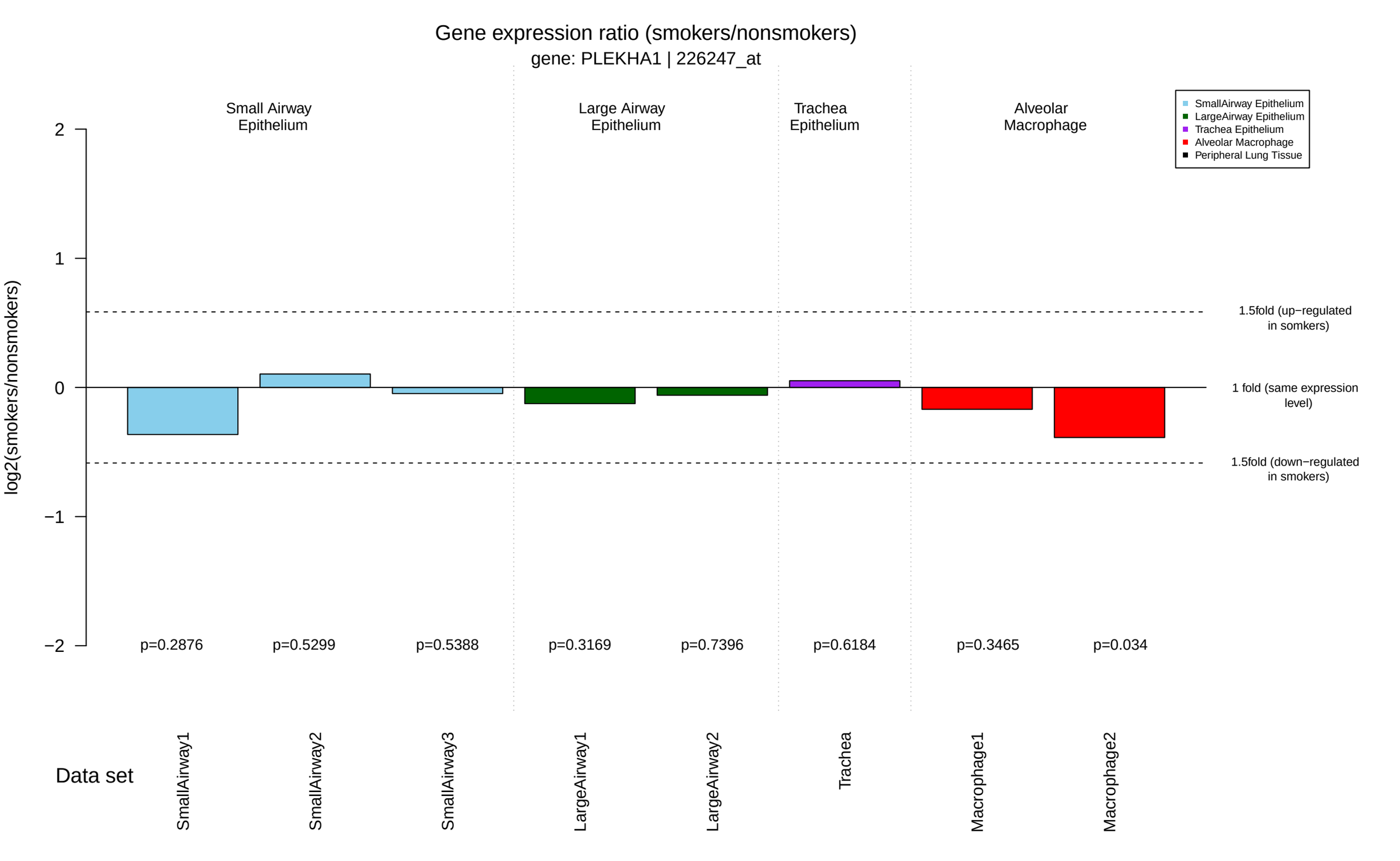
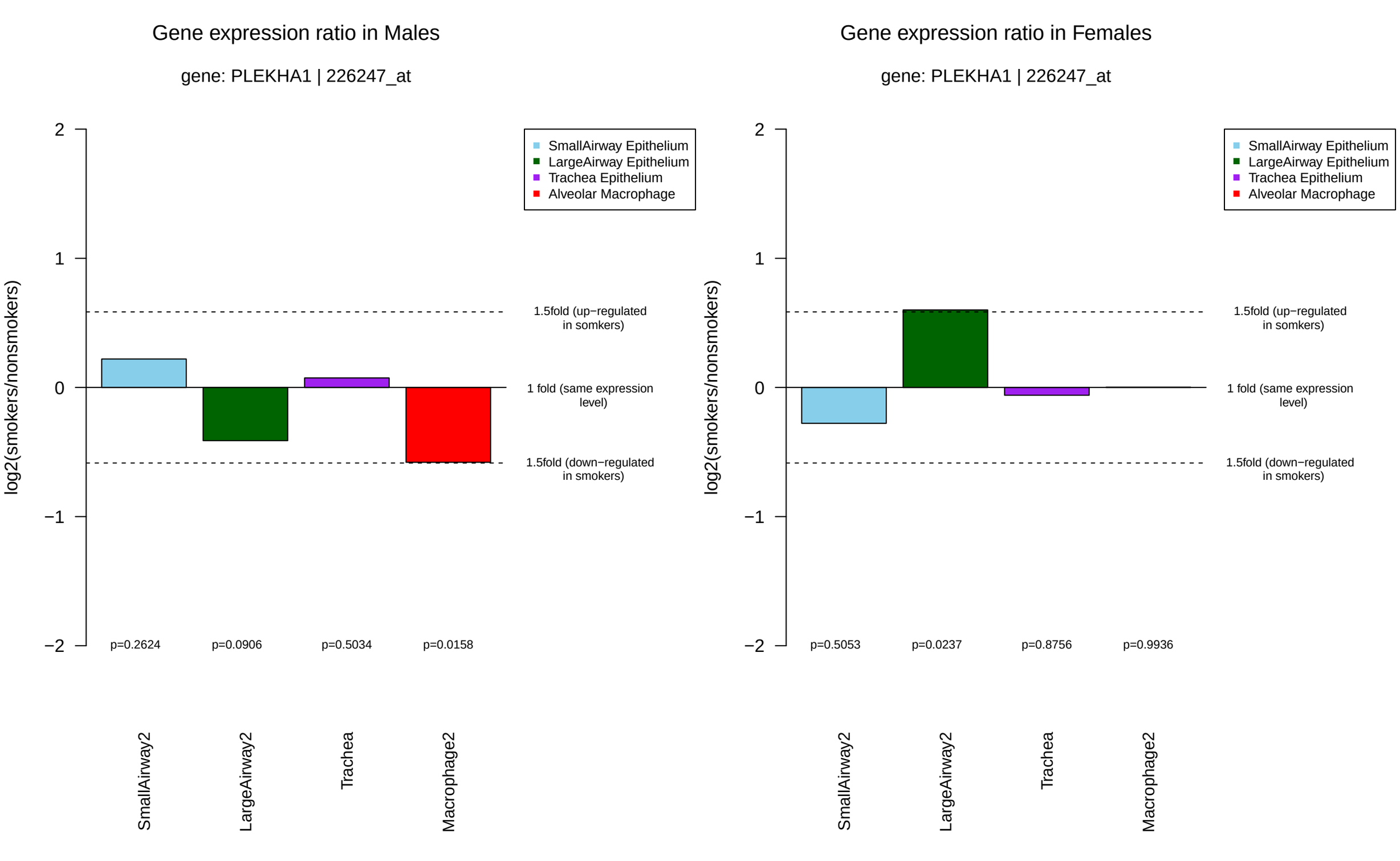
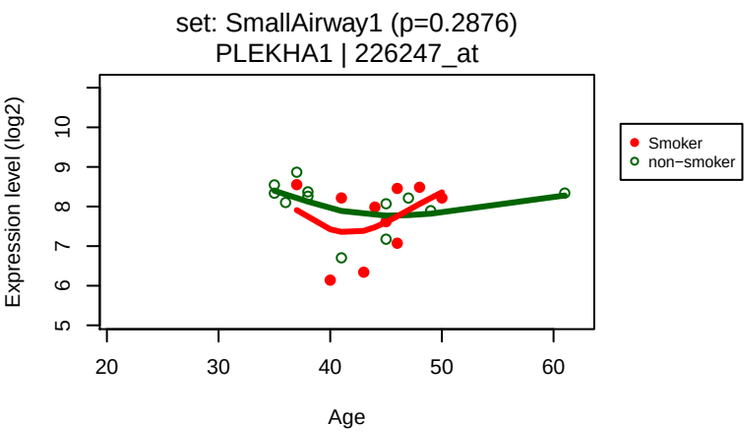
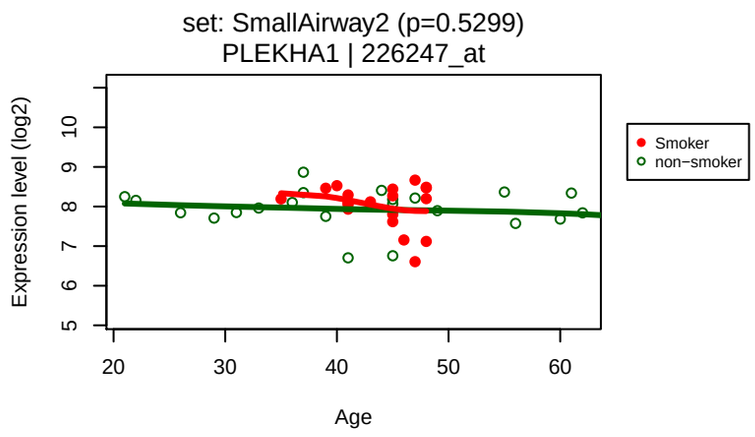
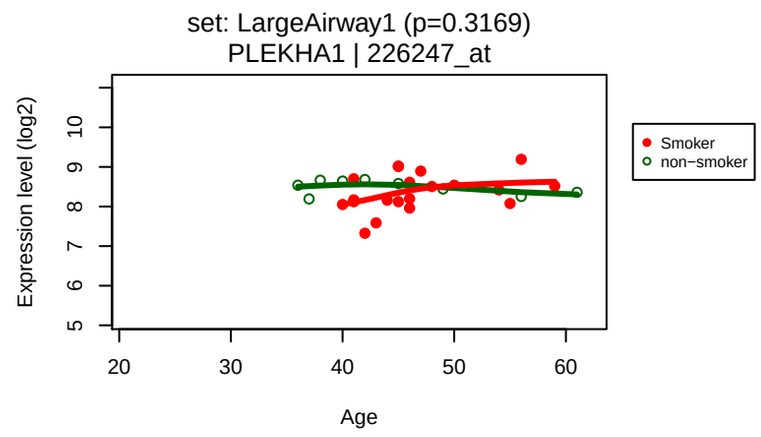

Gene expression of smokers and nonsmokers
gene: PLEKHA1 | 226247_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2876
p=0.5299
p=0.5388
p=0.3169
p=0.7396
p=0.6184
p=0.3465
p=0.034
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

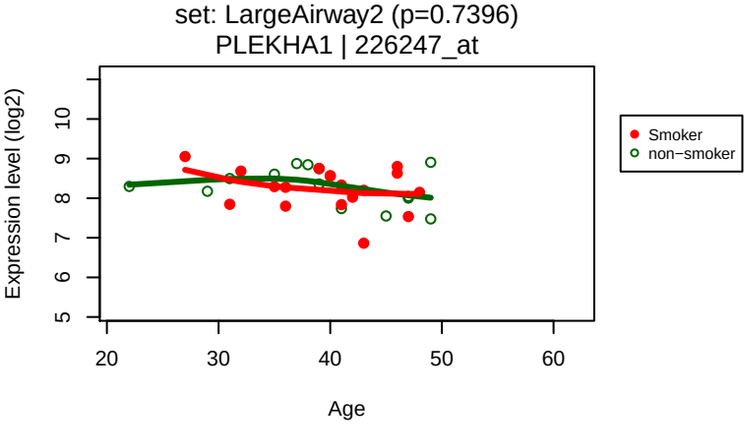

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA1 | 226247_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2876
p=0.5299
p=0.5388
p=0.3169
p=0.7396
p=0.6184
p=0.3465
p=0.034
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

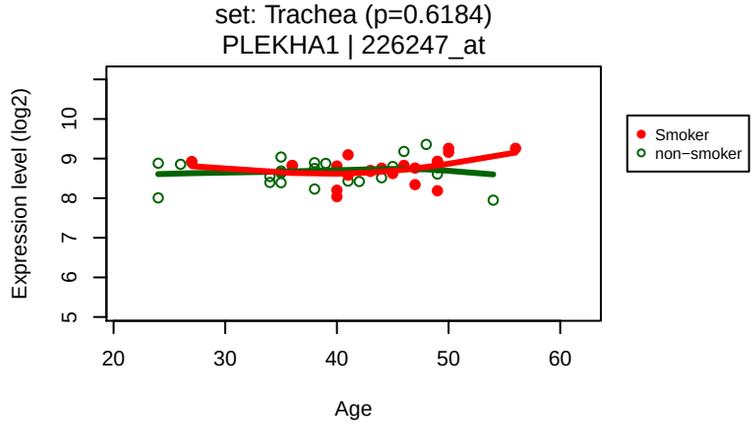

Gene expression ratio in Males
gene: PLEKHA1 | 226247_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2624
p=0.0906
p=0.5034
p=0.0158
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA1 | 226247_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5053
p=0.0237
p=0.8756
p=0.9936
SmallAirway2
LargeAirway2
Trachea
Macrophage2

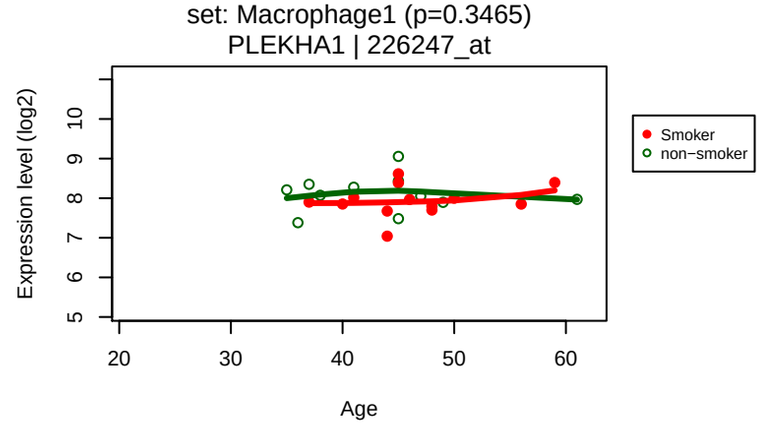

set: SmallAirway1 (p=0.2876)
PLEKHA1 | 226247_at
Expression level (log2)
Age
Smoker
non−smoker

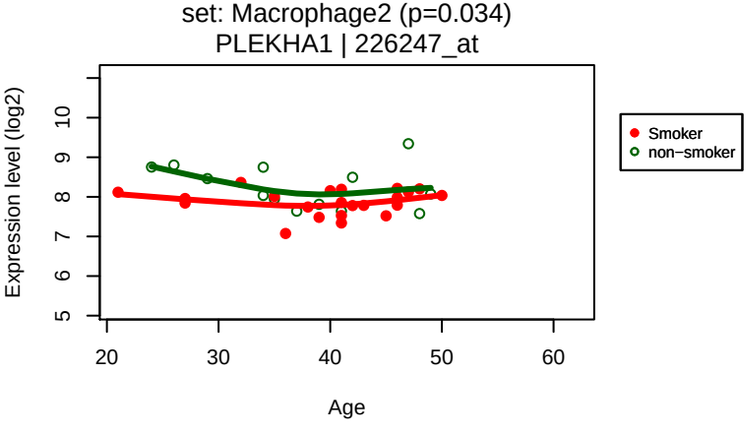

set: SmallAirway2 (p=0.5299)
PLEKHA1 | 226247_at
Expression level (log2)
Age
Smoker
non−smoker

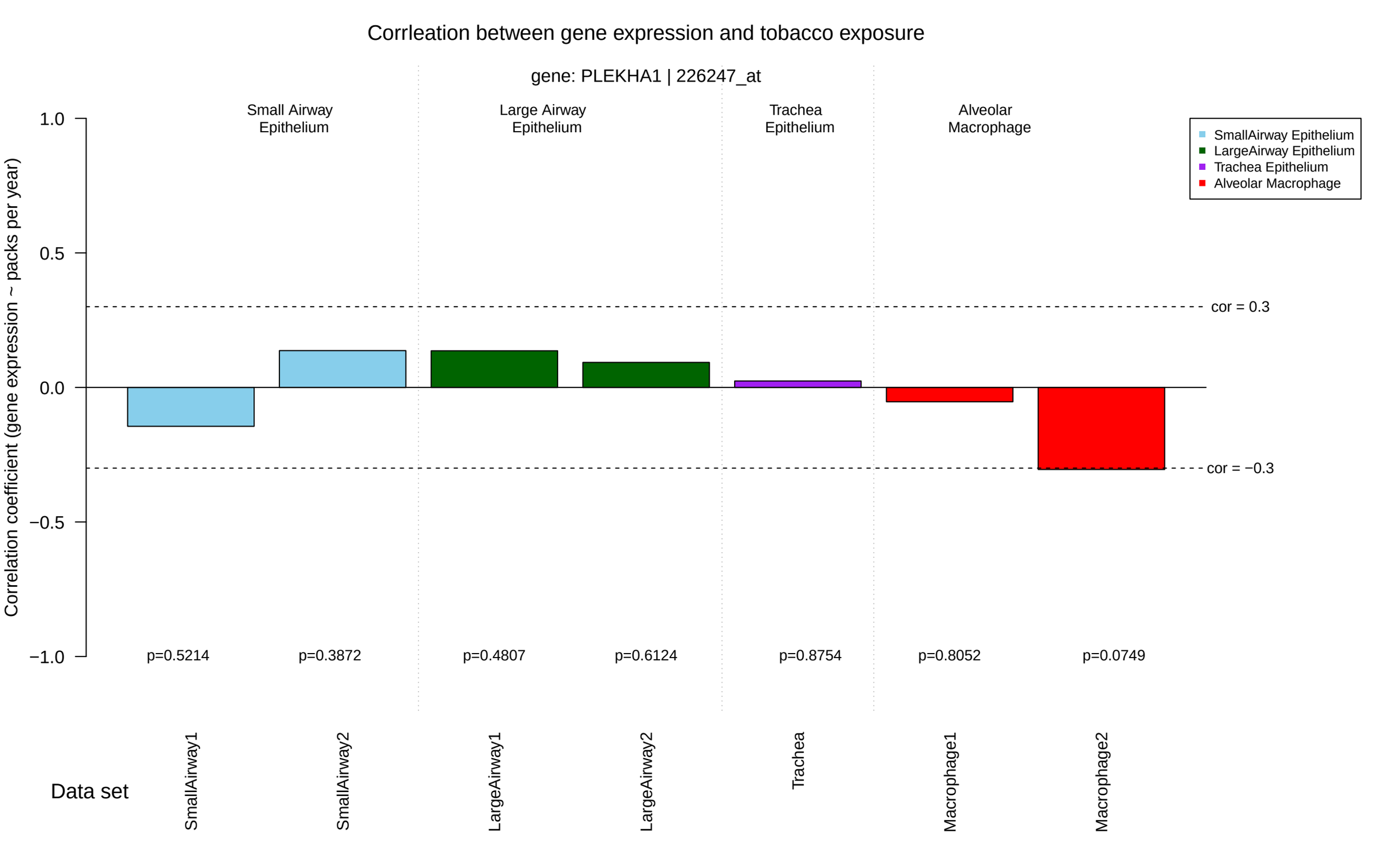

set: LargeAirway1 (p=0.3169)
PLEKHA1 | 226247_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7396)
PLEKHA1 | 226247_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6184)
PLEKHA1 | 226247_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3465)
PLEKHA1 | 226247_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.034)
PLEKHA1 | 226247_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA1 | 226247_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5214
p=0.3872
p=0.4807
p=0.6124
p=0.8754
p=0.8052
p=0.0749
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2